**DO NOW — RECALL**

# A*: heuristic search I

Name: _____ Class: _____ Date: _____

**1. Why are predecessors recorded when a distance improves?**

**2. Can Dijkstra assume negative edge weights are safe?**

**3. What is 2+3+2?**

**4. Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

They allow reconstruction of the actual shortest route.

### 2. Model answer

No. Its shortest-path guarantee requires non-negative weights.

### 3. Model answer

7.

### 4. Model answer

6.

### Vocabulary — know these terms

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

admissible heuristic — An estimate that never exceeds the true cheapest remaining cost.

NEW LEARNING

# Read, annotate and explain

### g, h and f have different meanings

g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

### Start with a frontier

From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

### Improve a discovered route

From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

### Stop on the selected goal

When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

### Keep the heuristic safe

The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace A* on P, Q, R, G using the supplied admissible estimates.

### Predict the next state

From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.

### Update the table

Through Q, R improves to g4,h2,f6; G gets g11.

### Update the table

From R, G improves to g6,h0,f6.

### Update the table

Select G: predecessor chain yields P→Q→R→G, cost 6.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

A node has g=7 and h=3. Give f and define each value.

### Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A* chooses the next vertex solely by its smallest heuristic h(n).**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.